# MILITARY SPECIFICATION

## WELDING SYSTEMS, METAL, PLASMA ARC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type metal welding system that utilizes a direct current (DC) power supply and the ionization of appropriate gases to produce a plasma welding arc.

1.2 Classification. Welding systems covered by this specification shall be of the following sizes, as specified (see 6.2.1):

TABLE I

Sizes and Characteristics

| Size | Power Output Range, Amperes | | | |
|---|---|---|---|---|
| | 100% Duty Cycle * | | 60% Duty @cle * | |
| | Minimum (Not More Than) | Max imum (Not Less Than) | Minimum (Not More Than) | Maximum (Not Less Than) |
| 1 | 0.1 | 10 | 0.1 | 15 |
| 2 | 0.1 | 75 | 0.1 | 100 |
| 3 | 3.0 | 80 | 3.0 | 100 |

*Duty Cycle based on 10-minute intervals.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.



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SPECIFICATIONS

Military

| | |
|---|---|
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-130 | Identification Marking of U. S. Military Property |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

| | |
|---|---|
| Handbook H28 | Screw Thread Standards for Federal Services |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1 and 6.3), the supplier shall furnish one complete welding system of each size specified in the contract or order for first article inspection and approval in accordance with Section 4 herein.

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3.2 Design. The plasma arc welding systems covered by this specification shall be new and of the manufacturer's current design. The design shall provide for convenient, efficient, and safe oper;tion. All parts subject to wear or requiring adjustment or maintenance shall be easily accessible, and disassembly shall not require the use of special tools other than those furnished by the manufacturer as part of the system.

3.2.1 Safety devices. All parts which are of such nature or so located to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. All electric circuits shall be fused as required to protect component parts. Safety devices shall not impair the performance of the equipment. When special safety devices are required, they shall be as specified (see 6.2.1).

3.2.2 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or system.

**3.2.3** Material. All material used in the manufacture of the welding systems shall be new and of the quality necessary to produce a unit to meet the requirements described herein.

3.3 Construction. The plasma arc welding systems shall be constructed so that when installed and connected to the specified power sources, they can be used for any purpose for which they were designed. The plasma arc welding systems shall be constructed of parts which are sound in structure and free from repair.

**3.3.1** Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. In no event shall processes such as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts.

3.3.2 Welding, brazing, or solderin~. Welding, brazing, or soldering shall be employed only where these operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

**3.3.3** Fastenin? devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Such fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.



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3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean'ed and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. Unless otherwise specified herein, external surfaces shall be finished in accordance with the manufacturer's standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), painting of the system shall be in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All screw threads shall be in the inch system and shall conform to the requirements of NBS Handbook H28.

3.4 Performance. The plasma arc welding systems shall have an adjustable power output range between the limits listed in 1.2. The systems shall be capable of operating at rated maximum power output without exceeding the designed operating characteristics of any component. When connected to the proper utility sources, and operated in accordance with the manufacturer's printed instructions, the plasma arc welding systems shall perform satisfactory welds in square butt joints in metals of the following thicknesses:

Size 1 - 0.005" to 0.050"
Size 2 - 0.005" to 0.186"
Size 3 - 0.005" to 0.250"

3.4.1 Special performance and requirements. When required, the plasma arc welding systems shall perform keyhole welds, and shall meet other special performance requirements as specified (see 6.2.1).

3.5 Components. Each plasma arc welding system shall consist primarily of a power supply unit, controls, torch, cooling water system, foot switch for welding arc power, and interconnecting cables and hoses.

3.5.1 Power supply unit and controls. The power supply unit and controls shall be contained within cabinets fabricated of metal, and constructed to a level of rigidity that will assure protection to enclosed components in production environments. All components shall be compatible with input power source and characteristics as specified (see 6.2.1). The system shall include the power supply components required for direct current output in both the transferred arc and the non-transferred arc modes. Arc

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welding current shall be adjustable within the specified power output range on a repetitive basis through a control on the front of the unit. An ammeter, scaled for the maximum r'ange to indicate arc current, shall be mounted on the front of the enclosed unit. Flowmeters, accurate to within ~ 5 per cent of full scale reading, shall be provided for metering the argon plasma gas, argon shielding gas, and mixtures of argon-hydrogen and argon-helium shielding gases. The flowmeters shall be scaled in standard cubic feet per hour (SCFH) or per cent, and shall operate with input pressure of between 20 and 40 PSI. A two-second time delay shall be provided for pre-flow and post-flow of the shielding gases. A switch or device shall be included in the control circuit to shut off the welding power in the event of a loss in the supply of cooling water. All controls shall be legibly and individually identified to their functions.

**3.5.2** Torch. The torch shall be designed for a continuous duty cycle rating at the maximum specified DC straight polarity current in the transferred arc mode. The torch shall be water cooled, and the torch head shall be angled for weld visibility. The torch shall be fitted with the manufacturer's standard tungsten electrode for the rating specified. Replacement of the electrode shall be accomplished with tools and gages normally furnished by the manufacturer. Means shall be provided for adjusting electrode setback. The electrode shall be self-centering in the torch orifice. The torch body shall provide separate channels for the flow of pla&na and shielding gases. The torch nozzle shall be fitted with replaceable gas cups or other means which shall provide for laminar flow of the shielding gases. Pilot arc starting shall be accomplished by the use of a high frequency discharge or by advancing and retracting the electrode. When specified (see 6.2.1), a torch holder shall be furnished.

**3.5.3** Cooling water system. Unless otherwise specified (see 6.2.1), the cooling water system shall include an electric motor driven pump capable of providing the flow rate and pressure required for 100 per cent duty cycle. When specified (see 6.2.1), a self-contained recirculating water supply system shall be furnished. The recirculating system shall consist of an electric motor driven pump, a reservoir, and a heat exchanger. Both cooling water systems shall be equipped with a pump that is of a type, or the system shall contain components, which permit pumping against a closed head without damage to the pump. Both cooling water systems shall operate on the input power source specified (see **3.5.1).**



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**3.5.4** Foot switch. The foot switch shall be of the swinging pedal type. The switch shall provide operator control for turning the welding power on and off. When specified (see 6.2.1), the foot switch shall also vary the welding power between the minimum and maximum ampere values for the size required in 1.2. The switch shall be connected to the control unit with a cable not less than 12 feet in length.

3.5.5 Interconnecting cables and hoses. All interconnecting cables and hoses with fittings attached shall be furnished, as well as the items necessary for connecting the unit to the specified utility sources. Work leads and hoses from the control unit to the torch shall be not less than 12 feet in length. Hoses shall be of the abrasion resistant type.

**3.6** Standard equipment. Unless otherwise specified (see 6.2.1), the unit shall be furnished with all standard equipment normally furnished by the supplier including any special tools or wrenches normally furnished.

3.7 Repair parts. When required, such repair parts as are specified (see 6.2.1), shall be furnished.

3.8 Electromagnetic interference control. When specified (see 6.2.1), the welding system shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-46L.

**3.9** Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistance metal nameplate marked in accordance with MIL-STD-130 shall be securely attached to the control unit. The nameplate shall contain the information listed below. If the system is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Welding power range, amperes
Manufacturer's serial number
Power input characteristics and ratings
Date of manufacture
Contract or order number
Federal Stock Number or Plant Equipment Code
u s

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3.10 Identification plate. When specified (see 6.2.1), a separate identification plate of corrosion resistant metal permanently and legibly marked with the caption "Departmental or Activity Identification" shall be securely attached to the control unit. The plate shall have blank space for the Government to stamp 25 letters or numbers of the same size as the caption.

3.11 Workmanship. Workmanship shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

## 4. CJJALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or one or more first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination, the tests, or the inspection shall be cause for rejection.



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4.5 Examination. The plasma arc'welding system shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The plasma arc welding system shall be connected to the specified utility sources and operated for a minimum of 30 minutes, during which time the controls and associated components will be actuated and observed to determine if the system operates properly.

**4.6.2** Special tests. When specified (see 6.2.1), the supplier shall perform special tests to indicate that the equipment complies with unusual operating conditions or optional equipment requirements as required.

**4.6.3** Electroma~netic interference control test. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control shall be tested in accordance with the applicable requirements of MIL-STD-461 to determine compliance with 3.8.

**4.6.4** Dielectric test. All dielectric materials operating at a difference in voltage of more than 25 volts shall be tested for leakage to ground and other conductors, by applying twice the operating voltage plus 1000 sine-wave rms volts continuously for one minute.

4.6.5 Weld test. The plasma arc welding system shall be tested, utilizing the following welding operations:

a. Fuse welding the minimum and maximum specified thickness AISI Type 301 and 304 stainless steel, using a flanged butt joint on the minimum thickness and a square butt joint on the maximum thickness. Direct current straight polarity power, shielding gas mix, flow rate, and welding speed shall be as standard with the manufacturer.

b. Fuse welding the minimum and maximum specified thickness alpha alloy grade commercially pure titanium material, using a flanged butt joint on the minimum thickness and a square butt joint on the maximum thickness. Direct current straight polarity power, shielding gas mix, flow rate, and welding speed shall be as standard with the manufacturer.



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c. Spot fuse welding of a 0.D30 inch diameter wire, AISI composition 316 stainless steel material, both cross and lap joints. Direct current straight polarity power, shielding gas mix, flow rate, and arc duration shall be as standard with the manufacturer.

The welds shall be free of cracks, porosity, voids, and structural defects, and shall be examined under 10 power magnification to determine quality of welds.

**4.7** Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging. packing. and markin~. Preservation, packaging, and packing shall be in accordance with Level A, B, or C of MIL-E-17555 as specified (see 6.2.1), and marking shall be as specified therein.

## 6. NOTES

**6.1** Intended use. The plasma arc welding systems covered by this specification are intended for fusion welds in thin metals and small diameter wires.

**6.2** Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. Special safety devices, if required (see 3.2.1).

e. Painting, if different (see 3.3.5).

f. Special operating performance and requirements, if required (see 3.4.1).



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13″ Input power source characteristics (see 3.5.1).

h. Torch holder, if required (see 3.5.2).

i. Cooling water system, if different (see 3.5.3).

j. Self-contained recirculating water system, if required (see 3.5.3).

k. Variable power foot switch, if required (see 3.5.4).

1. Standard equipment, if different (see 3.6).

m. Repair parts, when required (see 3.7).

n. Electromagnetic interference control, if required (see 3.8).

o. Nameplate, if different (see 3.9).

p″ Identification plate, if required (see 3.10).

~. First article inspection, if different (see 4.3).

r. Quality conformance inspection, if different (see 4.4).

s. Special tests, if required (see 4.6.2).

t. Electromagnetic interference control test, if different (see 4.6.3).

u. Level of preservation, packaging, and packing required (see 5.1).

6.2.2 Contract data requirements. Technical data such as the manufacturer's standard operators manual, parts list, and wiring diagrams will usually be required for delivery in connection with this specification. When technical data is required, it will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.



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**6.3** Instructions for first article approval. when first article inspection and approval is required for equipment furnished under this specification, the solicitation should conform to the applicable requirements of Section 1, Part 19 of the Armed Services Procurement Regulation.

Custodians:
Army - Wc
Navy - SH
Air Force - 84

Review Activities:
Army - EL, WC
Navy - SH
Air Force - 84

User Activities:
Army - SM, MU
Navy - MC

Preparing Activity:
DSA- 1P

Project Num her:
3438-0010



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

**INSTRUCTIONS:** This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANV 5PECIFICATII>M RE$2U!REMENT CO NSIOCREO 100 RIGIO

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REM&RKS (A,,.wh ~,,w, !!,, en, d., t,, .,!, ii'h <n",. be ,>( U*CI t" Itnfwo,,lql thin .pecfllcatlon. I( there ~~ ad[tlt~onal Papers ● ftach to lo?m ● nd place both In an .IBV.:CLM H,fdresmad to fwetJJwln# -cIIv)IY)

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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